Applied Mathematics Workshop 2026: Partial Differential Equations, Stochastic Control and Machine Learning

University of Economics Ho Chi Minh City (UEH), July 20-21, 2026

Location: Campus B, University of Economics Ho Chi Minh City
B1-205, 279 Nguyen Tri Phuong, Dien Hong Ward, Ho Chi Minh City, Vietnam

Organizing Committee

- **Huy Chau**, The University of Manchester, UK
- **Nhan-Phu Chung**, UEH, Vietnam
- **Xuan Truong Le**, UEH, Vietnam
- **Thai Nguyen**, Université Laval, Canada
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- **Ha Quyen Tran**, UEH, Vietnam

PROGRAM

Monday, July 20, 2026

Morning

	Poster Exhibition will be available throughout the day
07:30 – 08:30	Registration
08:30 – 08:40	Opening remarks
08:40 – 08:45	Photo
08:45 – 09:35	**Arnulf Jentzen** (The Chinese University of Hong Kong, Shenzhen & University of Münster) *Adam symmetry theorem: characterization of the convergence of the Adam optimizer*
9:35 – 10:10	**Nguyen Anh Dao** (University of Economics Ho Chi Minh City) *Uniqueness and quantitative behavior of solutions to aggregation equation with the Newtonian potential*
10:10 – 10:40	Tea break
10:40 – 11:10	**Ho Minh Duy Nguyen** (Max Planck Research School for Intelligent Systems & University of Stuttgart) *From Multimodal Foundation Models to Physical AI: Scalable and Structured Representation Learning*
11:10 – 11:45	**Viet Anh Nguyen** (Chinese University of Hong Kong) *Gelbrich Robust Sharpe Ratio Portfolio Optimization*
11:45 – 13:30	Lunch

Afternoon

13:30 – 14:20	**Hung Tran** (University of Wisconsin-Madison) *Nonexistence of vanishing-viscosity limits for mechanical Hamiltonian ergodic problems*
14:20 – 14:55	**Tan Nguyen** (National University of Singapore) *Steering Large Language Models: Geometric, Control-Theoretic, and Learning-Theoretic Foundations*
14:55 – 15:25	Tea break
15:25 – 15:55	**Dang-Khoa Nguyen** (University of Science, Vietnam National University Ho Chi Minh City) *Numerical algorithms for solving monotone inclusions from continuous to discrete settings*
15:55 – 16:30	**Nicolas Langrené** (Beijing Normal-Hong Kong Baptist University) *Scalable method for mean field control with kernel interactions via random Fourier features*
16:30 – 17:05	**Yeoneung Kim** (Seoul National University of Science and Technology) *Beyond the curse of dimensionality: Scalable policy iteration for HJB equations via neural operators*
18:30 – 21:00	Welcome dinner

Tuesday, July 21, 2026

Morning

07:30 – 11:00	Posters session
07:30 – 8:30	Registration
08:30 – 09:20	**Vuong Phan** (University of Southampton) *Faster than Fast-LTS: Robust Regression and Outlier Detection with DC Programming*
9:20 – 9:50	**Minh Thang Do** (The Chinese University of Hong Kong, Shenzhen) *Non-convergence to global minimizers in data driven supervised deep learning: Adam and stochastic gradient descent optimization provably fail to converge to global minimizers in the training of deep neural networks with ReLU activation*
9:50 – 10:20	Tea break
10:20 – 10:50	**Truong Cao Huynh** (Ho Chi Minh University of Education) *Besov regularity for weak solutions to a singular parabolic double-phase equation*
10:50 – 11:25	**Kyunghyun Park** (Nanyang Technological University) *Robust Q-Learning for mean field control uder Wasserstein uncertainty in common noise*
11:25 – 12:00	Closing Ceremony and Poster Awards
12:00 – 13:00	Lunch
14:00 – 21:00	Field trip

ABSTRACT

(1) **Uniqueness and quantitative behavior of solutions to aggregation equation with the Newtonian potential**

Nguyen Anh Dao, University of Economics Ho Chi Minh City, Vietnam

In this talk, we want to study solutions of the following problem:

$$\begin{cases} u_t = \operatorname{div}\left(u\nabla(-\Delta)^{-1}u^m\right) & \text{in } \mathbb{R}^N \times (0,T), \\ u(x,0) = f(x) & \text{in } \mathbb{R}^N, \end{cases}$$

with $m \geq 1$. We first establish the well-posedness theory for nonnegative densities $f(x)$ in $L^\infty_c(\mathbb{R}^N)$ (the space of bounded functions with compact support).

Concerning the qualitative behavior of solutions, we prove that the solution satisfies a universal bound

$$u(x,t) \leq (mt)^{-\frac{1}{m}}, \quad \text{for } (x,t) \in \mathbb{R}^N \times (0,\infty).$$

This implies the L^p-decay estimate of solution $1 < p < \infty$.
In addition, we investigated the asymptotic profile of u when $t \to \infty$. Precisely, for any $q \in [1,\infty)$, we have

$$\|u(t) - W(t)\|_{L^q(\mathbb{R}^N)} \lesssim t^{-\frac{q-1+2^{1-N}}{qm}}, \quad t > 0,$$

where $W(x,t)$ is the vortex patch solution.

Finally, we study the $\gamma-$Hölder regularity $\gamma \in (0,1)$ for solutions. Specifically, given $f \in C^\gamma(\mathbb{R}^N)$, we show that $u \in L^\infty(0,T;C^\gamma(\mathbb{R}^N))$ for $T > 0$. Moreover, the regularity is sharp in the sense that we cannot not obtain the $\gamma-$Hölder regularity for bounded solutions.
Hence, we extended the known results of the case $m = 1$ in the literature.

(2) **Non-convergence to global minimizers in data driven supervised deep learning: Adam and stochastic gradient descent optimization provably fail to converge to global minimizers in the training of deep neural networks with ReLU activation**

Minh Thang Do, The Chinese University of Hong Kong, Shenzhen, China

Deep learning (DL) methods – consisting of a class of deep neural networks (DNNs) trained by a stochastic gradient descent (SGD) optimization method – are nowadays key tools to solve data driven supervised learning problems. Despite the great success of SGD methods in the training of DNNs, it remains a fundamental open problem of research to explain the success and the limitations of such methods in rigorous theoretical terms. In particular, even in the standard setup of data driven supervised learning problems, it remained an open research problem to prove (or disprove) that SGD methods converge in the training of DNNs with the popular rectified linear unit (ReLU) activation function with high probability to global minimizers in the optimization landscape. In this work we answer this question negatively by proving that it does not hold that SGD methods converge with high probability to global minimizers of the objective function. Even stronger, in this work we prove for a large class of SGD methods that the considered optimizer does with high probability not converge to global minimizers of the optimization problem. It turns out that the probability to not converge to a global minimizer converges at least exponentially quickly to one as the width of the first hidden layer of the ANN (the number of neurons on the first hidden layer) and the depth of the ANN (the number of hidden layers), respectively, increase to infinity. The general non-convergence results of this work do not only apply to the plain vanilla standard SGD method but also to a large class of accelerated and adaptive SGD methods such as the momentum SGD, the Nesterov accelerated SGD, the Adagrad, the RMSProp, the Adam, the Adamax, the AMSGrad, and the Nadam optimizers. However, we would like to emphasize that the findings of this work do not imply that SGD methods do not succeed to train DNNs: it may still very well be the case that SGD methods provably succeed to train DNNs in data driven learning problems but, if this does hold, then the explanation for this can, due to the findings of this work, not be the convergence to global minimizers but instead can only be the convergence of the empirical risk to strictly suboptimal empirical risk levels, which should then be in some sense not too far away from the optimal empirical risk level (from the infimal value of the empirical risk function).

(3) **Adam symmetry theorem: characterization of the convergence of the Adam optimizer**
Arnulf Jentzen, The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen), China & University of Münster, Germany

In the training of artificial intelligence (AI) systems, often not the standard gradient descent (GD) method is the employed optimization scheme but instead suitable accelerated and/or adaptive GD methods – such as the momentum and the RMSprop methods – are considered. The most popular of such accelerated/adaptive optimization methods is presumably the Adam optimizer due to Kingma & Ba (2014). In this talk we introduce the notion of the stability region for general deep learning optimization methods and we reveal that among standard GD, momentum, RMSprop, and Adam we have that Adam is the only optimizer that achieves the optimal higher order convergence speed and also has the maximal stability region. In another main result of this talk, which we refer to as Adam symmetry theorem, we show for a simple class of quadratic stochastic optimization problems (SOPs) that Adam converges, as the number of Adam steps increases, to the solution of the SOP (the unique minimizer of the strongly convex objective function) if and only if the random variables in the SOP (the data in the SOP) are symmetrically distributed. In particular, in the standard case where the random variables in the SOP are not symmetrically distributed we disprove that Adam converges to the minimizer of the SOP as the number of Adam steps increases. The talk is based on joint works with Steffen Dereich, Thang Do, Robin Graeber, Sebastian Kassing, Adrian Riekert, and Philippe von Wurstemberger.

(4) **Besov regularity for weak solutions to a singular parabolic double-phase equation**
Cao Truong Huynh, Ho Chi Minh University of Education, Vietnam

This talk investigates the local Besov regularity for weak solutions to a class of degenerate parabolic double-phase equations characterized by non-standard (p, q)-growth conditions. The modulating coefficient $a(x, t)$ and the lead coefficient $b(x, t)$ are assumed to be spatial Hölder continuous, with $a(x, t)$ being strictly positive to ensure the stability of the double-phase structure. We establish fractional differentiability results for a non-linear auxiliary function $\Psi(z, \nabla u)$, which captures the intrinsic scaling of the multi-phase operator. By combining the finite difference method with recent Calderón-Zygmund estimates, we establish the optimal regularity $B^{\alpha}_{2,\infty,\text{loc}}(\Omega_T)$ for the homogeneous equation and a sharp transmission order of $\frac{\alpha q'}{2}$ for the non-homogeneous case. This approach effectively reconciles the space-time anisotropy and phase degeneracy inherent in the parabolic double-phase operator.

(5) **Beyond the curse of dimensionality: Scalable policy iteration for HJB equations via neural operators**

Yeoneung Kim, Seoul National University of Science and Technology, South Korea

This presentation introduces a novel, scalable mathematical framework to overcome the 'curse of dimensionality' in solving Hamilton-Jacobi-Bellman (HJB) equations for optimal control. By combining classical Policy Iteration (PI) with Physics-Informed Neural Networks (PINNs), the proposed mesh-free approach decomposes fully nonlinear HJB equations into a sequence of continuous linear PDEs. We provide rigorous theoretical guarantees, including geometric convergence, uniform boundedness, and the strict non-accumulation of L^2 neural approximation errors across finite-horizon, infinite-horizon (deterministic and stochastic), and entropy-regularized exploratory settings. Evaluated on complex tasks, including a 50-dimensional stochastic LQR problem and nonlinear benchmarks, our framework achieves rapid stabilization and monotonic performance improvements, successfully outperforming standard model-free reinforcement learning algorithms like SAC and PPO.

(6) **Scalable method for mean field control with kernel interactions via random Fourier features**

Nicolas Langrené, Beijing Normal-Hong Kong Baptist University, China

We develop a scalable algorithm for mean field control problems with kernel interactions by combining particle system simulations with random Fourier feature approximations. The method replaces the quadratic-cost kernel evaluations by linear-time estimates, enabling efficient stochastic gradient descent for training feedback controls in large populations. We provide theoretical complexity bounds and demonstrate through crowd motion and flocking examples that the approach preserves control performance while substantially reducing computational cost. The results indicate that random feature approximations offer an effective and practical tool for high dimensional and large scale mean field control.

(7) **Gelbrich Robust Sharpe Ratio Portfolio Optimization**

Viet Anh Nguyen, Chinese University of Hong Kong, China

Sharpe-ratio maximization is a foundational objective in portfolio allocation, but its empirical plug-in form is highly sensitive to estimation error in both expected returns and covariance matrices. We study a Gelbrich-robust Sharpe-ratio problem in which the unknown mean–covariance pair lies in a Gelbrich ambiguity ball around its nominal estimate. The resulting formulation is a max–min fractional optimization problem that jointly accounts for adversarial perturbations of mean returns and covariance matrix. We show that, under mild conditions, this problem admits an exact single-layer reformulation that could be solved to global optimality by a bisection algorithm with convex feasibility subroutines. Building on this exact bisection oracle, we introduce a data-driven ambiguity-ratio tuning scheme that normalizes the ambiguity radius across problem instances. To mitigate the computational complexity of repeated optimization, we propose a graph neural network to predict the optimal robust allocations. This network respects the permutation equivariance and scale invariance of the underlying problem, and it is trained in an unsupervised manner using a Karush-Kuhn-Tucker loss function. Experiments on equity universes compare the learned network with the exact oracle under temporal and dimensional distribution shifts, and evaluate its use for continuous ambiguity-ratio tuning.

(8) **From Multimodal Foundation Models to Physical AI: Scalable and Structured Representation Learning**

Ho Minh Duy Nguyen, Max Planck Research School for Intelligent Systems & University of Stuttgart, Germany

Multi-modal LLMs have transformed AI's ability to learn from rich, multi-modal inputs, spanning images, text, structured records, and more. Yet real-world domains such as healthcare and scientific discovery demand not only accuracy, but also scalability, robustness, and efficiency. This talk will present recent advances in scalable multi-modal learning with a focus on hybrid discrete–continuous representations that bridge structured knowledge and high-dimensional signals. We will discuss algorithmic designs that integrate combinatorial structures into neural networks via differentiable relaxations, enabling end-to-end training across heterogeneous modalities.

On the efficiency side, we will cover parameter-efficient fine-tuning and model/data compression strategies (e.g., token merging) that adapt large multi-modal foundation models to new domains at minimal computational cost. Applications will include medical vision–language models and the optimization and acceleration of large language models, illustrating how these techniques advance both predictive performance and interpretability while enabling deployment in resource-constrained settings.

Looking ahead, these ideas point toward Physical AI, where models must move beyond perception to generate actions consistent with the dynamics of the physical world. Emerging Vision–Language–Action (VLA) models aim to unify perception, reasoning, and control for embodied agents. We will discuss the key challenge of developing generative models that incorporate physical constraints and structured representations to enable reliable planning, manipulation, and interaction in complex real-world environments.

(9) **Numerical algorithms for solving monotone inclusions from continuous to discrete settings**

Dang-Khoa Nguyen, University of Science, Vietnam National University Ho Chi Minh City, Vietnam

We present an approach for finding zeros of a single-valued, monotone, and continuous operator via dynamical systems. We study the convergence behavior of the trajectory solutions, including explicit rates of decay for the operator norm. Numerical algorithms obtained by discretizing the continuous dynamics are also discussed. We also introduce the extensions of the results to more general monotone inclusion problems, where set-valued operators might be involved.

(10) **Steering Large Language Models: Geometric, Control-Theoretic, and Learning-Theoretic Foundations**

Tan Nguyen, National University of Singapore, Singapore

Inference-time activation steering enables lightweight behavioral control without weight updates, but its reliability depends on the mathematical structure of activation space. This talk develops three complementary perspectives. Geometrically, steering can be viewed as norm-preserving rotations in low-dimensional subspaces, whereas concepts are better understood as heterogeneous regions of a representation manifold rather than as single global directions. Control-theoretically, layerwise activations can be modeled as trajectories of a dynamical system, where feedback reduces drift, steady-state error, and overshoot. Finally, a learning-theoretic robustness view shows that high-dimensional representations make concepts linearly accessible but also expose exploitable directions. Together, these perspectives support predictable, tunable, and robust control of foundation models.

(11) **Robust Q-Learning for mean field control uder Wasserstein uncertainty in common noise**

Kyunghyun Park, Nanyang Technological University, Singapore

In this talk, I present a robust Q-learning algorithm for discrete-time mean-field control problems under Wasserstein uncertainty in the common noise law. The algorithm combines a quantization-and-projection scheme with a Wasserstein dual reformulation on the common-noise space. We establish its convergence together with finite-time iteration bounds for both synchronous and asynchronous learning schemes. Numerical experiments on systemic risk and epidemic models compare the asynchronous implementation with an idealized Bellman iteration, illustrate the robustness–performance tradeoff under common-noise misspecification, and report the observed convergence behavior of the asynchronous Q-learning algorithm. This is based on joint work with Mathieu Laurière and Ariel Neufeld, available as the preprint in arXiv:2606.20356.

(12) **Faster than Fast-LTS: Robust Regression and Outlier Detection with DC Programming**

Vuong Phan, University of Southampton, UK

When datasets contain outliers, robust regression is a well-established alternative to Ordinary Least Squares. A commonly employed robust estimator is Least Trimmed Squares (LTS), which computes the regression coefficients from a subset of observations. Determining the exact solution corresponds to a combinatorial problem with prohibitive computational costs, even for instances of moderate dimension. Thus, the most prevalent approach in practice remains a heuristic known as Fast-LTS. Although the heuristic often performs effectively, certain elements of the approach remain open to improvement. In particular, its core procedure provides robust results only when initialized with a large number of starting points. To address the heuristic's limitations, this paper reformulates the LTS problem as a concave minimization problem subject to a capped simplex constraint, and proposes the successive Boosted Difference of Convex Functions Algorithm (sBDCA) as a solution method. Theoretically, we establish via the Łojasiewicz property that sBDCA converges to a local solution with a linear rate in the fastest case. To ensure robustness from a single initialization in practice, we derive and integrate a problem-specific preconditioning matrix into the algorithmic setup. Building on this theoretical foundation, we conduct numerical studies on various synthetic and real-world datasets to demonstrate the effectiveness of sBDCA with preconditioning. Specifically, we show that our approach is up to 3.25 times faster than Fast-LTS and achieves up to 90

(13) **Nonexistence of vanishing-viscosity limits for mechanical Hamiltonian ergodic problems**

Hung Tran, University of Wisconsin-Madison, US

I will discuss the classical vanishing viscosity process for Hamilton-Jacobi and Burgers' equations. I'll then address the mechanical Hamiltonian ergodic problems and the open question stated by Jauslin—Kreiss—Moser, Evans, and many others. I will then give a negative answer to the question by constructing a one-dimensional example. Joint work with Ziran Liu and Yifeng Yu.